



Covariance Matrix Adaptation Evolution Strategy Algorithm in the Context of Seismic Inversion: Proposal, Application, and Analysis Compared to Ensemble Smoother with Multiple Data Assimilation

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ABSTRACT

Seismic reservoir characterization faces non-uniqueness challenges, requiring robust uncertainty quantification. This study evaluates the UH-CMA-ES algorithm against the ESMDA method using synthetic seismic data. Both methods effectively minimize seismic misfit. However, UH-CMA-ES provides significantly more focused uncertainty envelopes. Statistical analysis via the Kolmogorov-Smirnov test demonstrates that UH-CMA-ES can generate property distributions indistinguishable from the ground truth. While computationally more expensive and exhibiting a reduction in global coverage, UH-CMA-ES proves to be a high-fidelity alternative for precise local exploitation in ill-posed inversion problems.

KEYWORDS. UH-CMA-ES; ES-MDA; Seismic Inversion; Reservoir characterization; History Matching;

EN&OG - Operational Research in Energy, Oil, and Gas Areas; MH - Meta-heuristics; OC - Continuous Optimization;

1. Introduction

Seismic reservoir characterization plays a pivotal role in the oil and gas industry, providing critical insights for reservoir management and drilling decisions. However, it constitutes a highly complex inverse problem, facing inherent challenges due to non-uniqueness. Different subsurface geological configurations can yield the same seismic response, making deterministic solutions insufficient. (Grana et al. [2021]; Zhao et al. [2024]). Probabilistic approaches are crucial for quantifying these uncertainties, thereby guiding decision-making. With the increasing demand for high-resolution models, the efficiency and robustness of inversion methods become critical, especially in high-dimensional and highly non-linear problems.

Reservoir characterization through seismic inversion constitutes a classic inverse problem, aiming to estimate subsurface elastic properties—such as compressional and shear wave velocities (v_p and v_s) and density (ρ)—from seismic data observed at the surface. A particularity of this domain is the ill-posed nature of the problem, manifested in the coexistence of multiple mathematically equivalent solutions, a phenomenon known as non-uniqueness. In other words, different configurations of properties can produce simulated responses that fit the field observations equally well, attributing uncertainty to a meaning related to the space of possible models.

Advanced methods such as Ensemble Smoother(ES) based and the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) emerge as promising alternatives (Emerick and Reynolds [2013]; Ma and Bi [2019]; Liu and Grana [2018]). Ensemble Smoother with Multiple Data Assimilation (ES-MDA), a widely used ensemble-based data assimilation method in the field, is known for its ability to handle large volumes of data and its potential computational efficiency (Emerick [2025]). However, its effectiveness in uncertainty quantification can be sensitive to the ensemble size (Yang et al. [2023]). On the other hand, CMA-ES is an evolutionary optimization algorithm that is robust for complex and nonlinear problems (Hansen et al. [2008]), but it can present a high computational cost in high dimensions (Yang et al. [2023]).

Despite comparative studies in related fields (Yang [2024] in hydrology), this work proposes the use of the CMA-ES algorithm specifically tailored to the challenges of seismic reservoir inversion. This project is motivated by the need to investigate modifications to enhance the optimizer's robustness, the objective is to evaluate the performance of the adapted CMA-ES in terms of accuracy, efficiency, and reliability in uncertainty quantification, having ES-MDA as a baseline. CMA-ES was specifically chosen due to its derivative-free evolutionary nature, which makes it highly capable of navigating the severe non-linearity and complex multi-modal landscapes typical of seismic inversion, where traditional gradient-based methods often fail or get trapped in local minima.

The main contributions of this research are: (1) the adaptation of the UH-CMA-ES algorithm to handle the inherent non-uniqueness of seismic inversion; and (2) a comparative analysis demonstrating that UH-CMA-ES provides highly focused uncertainty envelopes compared to ES-MDA. The remainder of this paper is organized as follows: Section 2 details the theoretical framework of both methods. Section 3 presents the methodology and experimental design. Finally, Sections 4 and 5 discuss the results and conclude the paper, respectively.

2. Theoretical Framework

From a statistical inference perspective, seismic inversion is classically structured through a Bayesian formulation (Tarantola [2005]; Grana et al. [2021]). The nonlinear physical mapping that connects the vector containing the subsurface elastic or petrophysical models, $\mathbf{m}$, to the observed seismic data, $\mathbf{d}_{\text{obs}}$, is defined by the expression:

$$\mathbf{d}_{\text{obs}} = \mathbf{g}(\mathbf{m}) + \epsilon \quad (1)$$

In this context, $\mathbf{g}$ acts as the forward model operator, mapping the model space to the data domain. The vector ϵ quantifies acquisition uncertainties and numerical imperfections, conventionally modeled as a multivariate normal random variable with zero mean and a measurement error covariance matrix $\mathbf{C}_D$.

The probabilistic model update is based on Bayes' Theorem, which establishes that the posterior Probability Density Function (PDF), $P(\mathbf{m}|\mathbf{d}_{\text{obs}})$, is evaluated from the product of the prior state of knowledge, represented by the prior distribution $P(\mathbf{m})$, and the data likelihood function, $P(\mathbf{d}_{\text{obs}}|\mathbf{m})$:

$$P(\mathbf{m}|\mathbf{d}_{\text{obs}}) \propto P(\mathbf{m})P(\mathbf{d}_{\text{obs}}|\mathbf{m}) \quad (2)$$

By adopting the premise that both the measurement errors and the prior information follow a Gaussian behavior, the posterior probability density function becomes proportional to the exponential of a negative objective function. Consequently, the statistical challenge of finding the Maximum A Posteriori (MAP) estimate is mathematically strictly equivalent to solving a deterministic continuous least squares optimization problem. From the perspective of inverse theory, this formulation transforms the probabilistic inference into a Tikhonov regularization problem (Tartola [2005]), wherein the inverse of the spatial covariance matrix, $\mathbf{C}_M^{-1}$, acts as the regularization operator anchoring the solution to a reference prior model $\mathbf{m}_{\text{prior}}$ to stabilize the ill-posed minimization of the data misfit. This process is mathematically reduced to minimizing the following objective function:

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{g}(\mathbf{m}) - \mathbf{d}_{\text{obs}})^T \mathbf{C}_D^{-1}(\mathbf{g}(\mathbf{m}) - \mathbf{d}_{\text{obs}}) + \frac{1}{2}(\mathbf{m} - \mathbf{m}_{\text{prior}})^T \mathbf{C}_M^{-1}(\mathbf{m} - \mathbf{m}_{\text{prior}}) \quad (3)$$

In this formulation, the first term represents the primary goal of minimizing the data misfit, while the second term acts as a spatial regularizer anchoring the solution to the prior geological knowledge.

Due to the highly nonlinear nature of the operator $\mathbf{g}(\mathbf{m})$ and the presence of noise in the data, the minimization of the function $J(\mathbf{m})$ becomes a challenge for local gradient-based optimizers. In this context, stochastic and computational intelligence algorithms emerge as robust alternatives for exploring the model space. While ES-MDA addresses the problem through iterative ensemble assimilation, the adapted CMA-ES proposes an adaptive evolutionary strategy specifically designed to handle the noise and complexity inherent in these scenarios.

2.1. Uncertainty Handling CMA-ES

The objective of CMA-ES is to optimize nonlinear functions in continuous domains (Hansen and Ostermeier [2001]). The method assumes that parameters follow a multivariate normal distribution $\mathcal{N}(\mathbf{m}, \mathbf{C})$ and generates a population of candidate solutions $\mathbf{m}_j^{(l)}$ at each iteration based on the relationship:

$$\mathbf{m}_j^{(l+1)} \sim \mathbf{m}^{(l)} + \sigma^l \mathcal{N}(0, \mathbf{C}^{(l)}), \quad j = 1, 2, \dots, \lambda \quad (4)$$

Where the superscript (l) denotes the current iteration step, $\sigma^{(l)}$ is the step size and λ is the population size. The best μ solutions are selected to update the mean ($\mathbf{m}$) and the covariance matrix ($\mathbf{C}$). The covariance matrix and $\sigma^{(l)}$ adapt according to (Hansen and Ostermeier [2001]); thus, correlations from successive iterations are incorporated and the search radius is modified. Consequently, the number of required objective function evaluations is reduced and the population

spread is optimized. These iterative adaptations make CMA-ES particularly robust in maintaining convergence even with small populations or functions with many local optima.

However, the non-uniqueness of the problem makes the standard method unfeasible, as a single deterministic solution is insufficient for reservoir characterization. It is necessary to find a region of solutions rather than collapsing into a single optimum. Adding a stochastic nature to the objective function allows for the incorporation of non-uniqueness into the problem. Nevertheless, it is necessary to modify CMA-ES to handle this uncertainty and maintain a controlled spread of candidate solutions, generating an ensemble of valid models for the ill-posed problem.

The adaptation of CMA-ES to handle noisy or stochastic objective functions is called Uncertainty Handling CMA-ES (UH-CMA-ES) and was consolidated by works such as (Kruisselbrink et al. [2011]; Hansen et al. [2008]). It is based on the premise that the observed objective function $J_{\text{noise}}(\mathbf{x})$ is a random variable composed of the true function $J(\mathbf{x})$ and a stochastic error ϵ :

$$J_{\text{noise}}(\mathbf{x}) = J(\mathbf{x}) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2) \quad (5)$$

To mitigate the effect of ϵ on the population ranking, UH-CMA-ES utilizes an estimate of the mean fitness $\hat{J}(\mathbf{x})$ through n_{eval} independent re-evaluations of the same individual:

$$\hat{J}(\mathbf{x}) = \frac{1}{n_{\text{eval}}} \sum_{i=1}^{n_{\text{eval}}} J_{\text{noise},i}(\mathbf{x}) \quad (6)$$

The core of the algorithm proposed by Kruisselbrink lies in the rank-change criterion to decide the increase of n_{eval} . A subset of re-evaluated individuals is defined, and the ranking stability is verified. The stability condition is checked by comparing the fitness difference between two individuals $\mathbf{x}_i, \mathbf{x}_j$ and the combined uncertainty:

$$|\hat{J}(\mathbf{x}_i) - \hat{J}(\mathbf{x}_j)| < \alpha \cdot \sqrt{s_{\text{err}}(\mathbf{x}_i)^2 + s_{\text{err}}(\mathbf{x}_j)^2} \quad (7)$$

where:

$$s_{\text{err}}(\mathbf{x}) = \sqrt{\frac{1}{n_{\text{eval}}(n_{\text{eval}} - 1)} \sum_{i=1}^{n_{\text{eval}}} (J_{\text{noise},i}(\mathbf{x}) - \hat{J}(\mathbf{x}))^2} \quad (8)$$

In (7), α is a threshold parameter (typically $\alpha \in \mathbb{R}_+^*$) that regulates noise sensitivity. A larger α makes the algorithm more tolerant to noise before requesting re-evaluations, while a smaller α increases the strictness of the ranking stability. If this inequality is frequently satisfied, indicating that noise prevents a clear distinction between candidates, the algorithm increases the number of evaluations for the next iteration $l + 1$ following a proportion rule regulated by a real parameter:

$$n_{\text{eval}}^{(l+1)} = \lceil n_{\text{eval}}^{(l)} \cdot \beta \rceil \quad (9)$$

This mechanism ensures that the selection pressure of CMA-ES, based on recombination weights w_i , is applied to statistically distinguishable fitness values. The update of the distribution mean $\mathbf{m}^{(l+1)}$ and the covariance matrix $\mathbf{C}^{(l+1)}$ preserves their integrity even under high noise levels, allowing the step size $\sigma^{(l)}$ not to decay prematurely due to stochastic fluctuations of the objective function.

2.2. ES-MDA

ES-MDA originates from the Ensemble Smoother (ES) according to Emerick [2025], an algorithm that seeks to update an initial set of geological models (the prior), generated from geo-statistical knowledge, by assimilating all observed data globally in time and space. Mathematically, the update for each ensemble member j (where $j = 1, 2, \dots, N_e$) is given by:

$$\mathbf{m}_j^{(l+1)} = \mathbf{m}_j^{(l)} + \mathbf{C}_{MD} (\mathbf{C}_{DD} + \mathbf{C}_D)^{-1} (\mathbf{d}_{\text{noise},j} - \mathbf{g}(\mathbf{m}_j^{(l)})) \quad (10)$$

Where $\mathbf{m}_j^{(l)}$ represents the geological model vectors before (forecast/prior) and after (analysis/posterior) the assimilation, $\mathbf{g}(\cdot)$ is the nonlinear forward modeling operator, $\mathbf{C}_{MD}$ represents the cross-covariance matrix between the model variables and the ensemble-simulated responses, $\mathbf{C}_{DD}$ is the auto-covariance matrix of the simulated responses, and $\mathbf{C}_D$ is the covariance matrix of the seismic data measurement errors.

The addition of stochastic noise to the actual prior data ($\mathbf{d}_{\text{obs}}$) for each ensemble member is both a statistical and geophysical step. From a geophysical perspective, reflection seismic has inherent resolution limits and is inherently noisy; forcing all ensemble models to match a single deterministic observed data point ($\mathbf{d}_{\text{obs}}$) exactly would induce "geological overfitting," generating unrealistic properties to compensate for acquisition noise. By perturbing the data according to the variance $\mathbf{C}_D$, it is ensured that the final ensemble appropriately represents the posterior uncertainty space, accommodating the ambiguity of the problem.

ES-MDA introduces the idea of performing data assimilation iteratively, damping the impact of nonlinearity. The data fit does not occur all at once but is divided into N_a assimilation steps, to maintain correct Bayesian sampling, the data likelihood function is tempered (?). Mathematically, tempering the likelihood by a weight α_l is equivalent to inflating the data error covariance matrix ($\mathbf{C}_D$) at each step, such that the condition $\sum_{l=1}^{N_a} \frac{1}{\alpha_l} = 1$ is satisfied.

Thus, the iterative ES-MDA update equation at step l becomes:

$$\mathbf{m}_j^{(l+1)} = \mathbf{m}_j^{(l)} + \mathbf{C}_{MD}^{(l)} \left(\mathbf{C}_{DD}^{(l)} + \alpha_l \mathbf{C}_D \right)^{-1} (\mathbf{d}_{\text{noise},j}^{(l)} - \mathbf{g}(\mathbf{m}_j^{(l)})) \quad (11)$$

where the matrices $\mathbf{C}_{MD}^{(l)}$ and $\mathbf{C}_{DD}^{(l)}$ are recalculated at each iteration based on the new ensemble dynamics. The perturbed data vector at each iteration is also adjusted by the inflation:

$$\mathbf{d}_{\text{noise},j}^{(l)} = \mathbf{d}_{\text{obs}} + \sqrt{\alpha_l} \epsilon_j, \quad \text{with } \epsilon_j \sim \mathcal{N}(0, \mathbf{C}_D) \quad (12)$$

The introduction of the coefficient α_l serves a dual purpose: it stabilizes the matrix inversion by increasing diagonal dominance in $(\mathbf{C}_{DD}^{(l)} + \alpha_l \mathbf{C}_D)$ and preserves the ensemble uncertainty throughout the iterations. After N_a steps, ES-MDA provides a robust set of geological models that satisfy the observed seismic data while maintaining the necessary diversity for risk quantification in reservoir characterization.

3. Methodology

To evaluate the performance of the proposed UH-CMA-ES approach, a quantitative assessment was adopted using ES-MDA as a baseline on the inversion of synthetic seismic data. The use of a synthetic reference model (known as the ground truth or $\mathbf{m}_{\text{real}}$) is essential, as it allows for the exact measurement of the estimation error and the validation of the uncertainty coverage mapped by the algorithms.

The entire infrastructure for forward modeling, observed data, and the ES-MDA algorithm were utilized according to the open-source library SeReM (Seismic Reservoir Modeling) (Grana et al. [2021]). Only the UH-CMA-ES routine was custom-implemented, integrated with the SeReM operators in a GNU Octave environment.

3.1. Operationalizing UH-CMA-ES for Seismic Inversion

A critical aspect of operationalizing UH-CMA-ES for seismic inversion is adapting its output to represent an uncertainty space rather than a single deterministic optimum. In this study, instead of merely extracting the single best individual at the final iteration, the uncertainty ensemble is generated by extracting the final converged population of candidate solutions. This procedure ensures that the intrinsic non-uniqueness of the problem is captured, providing a valid ensemble of geological models for risk analysis.

In this paper, the fitness of each candidate model evaluated by UH-CMA-ES is governed by a stochastic objective function, based on the formulation presented by Yang et al. [2023]. This formulation quantifies the geological model mismatch by balancing the prediction error relative to the perturbed seismic data with a regularization term anchored in the geological prior information. Mathematically, the cost to be minimized for a given model $\mathbf{m}$ is expressed as:

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{d}_{\text{noise}} - \mathbf{g}(\mathbf{m}))^T \mathbf{C}_D^{-1}(\mathbf{d}_{\text{noise}} - \mathbf{g}(\mathbf{m})) + \frac{1}{2}(\mathbf{m} - \mathbf{m}_{\text{prior}})^T \mathbf{C}_M^{-1}(\mathbf{m} - \mathbf{m}_{\text{prior}}) \quad (13)$$

where the term $\mathbf{d}_{\text{noise}}$ represents the perturbed observation data vector.

3.2. Experimental Design and Forward Modeling

The experiment involves the inversion of seismic traces for the simultaneous estimation of subsurface elastic properties: compressional and shear wave velocities (v_p and v_s , typically measured in m/s or km/s) and density (ρ , typically measured in g/cm^3), which constitute the parameter vector $\mathbf{m}$.

To generate the perturbed data $\mathbf{d}_{\text{noise}}$ from the SeReM dataset, simulating real acquisition conditions and imposing the ill-posed nature of the problem, uncorrelated Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbf{C}_D)$ was added to the existing observed data in the dataset. The covariance matrix $\mathbf{C}_D$ was defined based on a pre-established signal-to-noise ratio (SNR). The prior information $\mathbf{m}_{\text{prior}}$ was constructed from a low-frequency trend model applied to the ground-truth properties $\mathbf{m}_{\text{real}}$ available in the same dataset.

3.3. Configuration of Inversion Algorithms

To ensure a fair comparison, both methods were initialized using the same prior probability distribution $\mathcal{N}(\mathbf{m}_{\text{prior}}, \mathbf{C}_M)$. Due to the differences in magnitude and variance between v_p , v_s , and ρ , the physical parameters were normalized to a standard space during optimization, stabilizing the calculation of the covariance matrices.

One realization of the multiple data assimilation method was executed using the native SeReM routine as a baseline, where an initial ensemble of size 100 was sampled via sequential geostatistical simulation, the number of assimilation steps was fixed at 4, and the inflation coefficients α_l were defined in a decreasing sequence to dampen geometric nonlinearity in the initial steps, respecting the strict constraint $\sum_{l=1}^{N_a} \alpha_l^{-1} = 1$.

Since the evolutionary algorithm possesses a stochastic nature, three independent realizations (denoted as Run 1, Run 2, and Run 3) were executed to evaluate the consistency and stability of the algorithm's convergence. The three runs were configured with a base population size λ given by $4 + \lfloor 3 \log(N) \rfloor$, as in Kruisselbrink et al. [2011]. The initial global step-size parameter, $\sigma^{(0)}$, was set to 1. The dynamic re-evaluation of the objective function n_{eval} operated within the integer range $[1, 50]$. The standard deviation associated with the perturbation was fixed at 10^{-4} , so the difference is only by their initial random seeds.

3.4. Performance Evaluation Metrics

The comparison between UH-CMA-ES and ES-MDA is based on three analytical axes aligned with the ill-posed nature of the problem:

1. **Accuracy of Fit (Data Space and Model Space):** The assessment of methodological adherence occurs on two fronts. In the data domain, the minimization of the misfit is analyzed through the angle-dependent seismic residuals (Near, Mid, and Far). In the parameter space, the accuracy of the estimates relative to the ground-truth model ($\mathbf{m}_{\text{real}}$) is quantitatively measured by the non-parametric Kolmogorov-Smirnov (KS) test. The test statistic (KS Stat) evaluates the maximum absolute distance between the cumulative distribution functions of the estimate and the true model, while the p-value assesses the statistical significance of the equivalence between these distributions.
2. **Uncertainty Quantification (Posterior Quality):** Statistical reliability is evaluated by observing the spread (uncertainty envelope) and the coverage rate (the frequency with which the true model is encompassed) of the ensembles generated by each method. This analysis verifies whether the estimated empirical standard deviation adequately covers the fluctuations of $\mathbf{m}_{\text{real}}$, mapping the trade-off between restricted exploitation in local basins of attraction and the risk of excessive variance collapse (ensemble collapse).
3. **Computational Efficiency:** Efficiency is measured not by absolute CPU time, but by the computational cost represented by the total number of calls to the forward operator $g(\cdot)$. Given the very high computational load imposed by wave propagation simulation in real scenarios, efficiency is not the computation time, but rather by the total number of calls to the forward operator. For ES-MDA, this cost is deterministic and linear: $N_{\text{sim}} = N_e \times N_a$, where N_e is the ensemble size and N_a is the number of assimilation steps. For UH-CMA-ES, the cost is cumulative and dynamic, depending on the re-evaluation history: $N_{\text{sim}} = \sum_l \sum_{i=1}^{\lambda} n_{\text{eval}}^{(l,i)}$.

4. Results

Both methods demonstrated the ability to satisfactorily minimize the seismic misfit. The residual analysis for the Near, Mid, and Far angles indicates that both ES-MDA and UH-CMA-ES (considering both the best individual and the ensemble mean) reached an equivalent convergence level. This behavior is evidenced in Figure 1, where the residual curves for both methods overlap, indicating that different models in the parameter space result in similar seismic responses.

Although the seismic fit is comparable, the distribution of the estimated geological models (v_p, v_s, ρ) reveals fundamental distinctions in how each algorithm maps uncertainty. UH-CMA-ES presented a visibly narrower uncertainty envelope compared to ES-MDA, a behavior directly reflected in the reduction of spread values (Table 1), which are approximately 45% lower for the v_p parameter in Run 3. This characteristic suggests that the adaptive evolution algorithm tended to concentrate the probability density in a more restricted region of the model space, refining the search for better physical parameters.

Given the ill-posed nature of seismic inversion, this behavior indicates that the optimizer located a robust solution within a basin of attraction, exploring it with greater local precision at the expense of the global diversity presented by ES-MDA. While ES-MDA acts as a smoother that preserves a broad sampling of the prior knowledge, UH-CMA-ES promotes focused convergence, resulting in an uncertainty estimate that is less conservative and more oriented toward the exploitation of high-likelihood solutions.

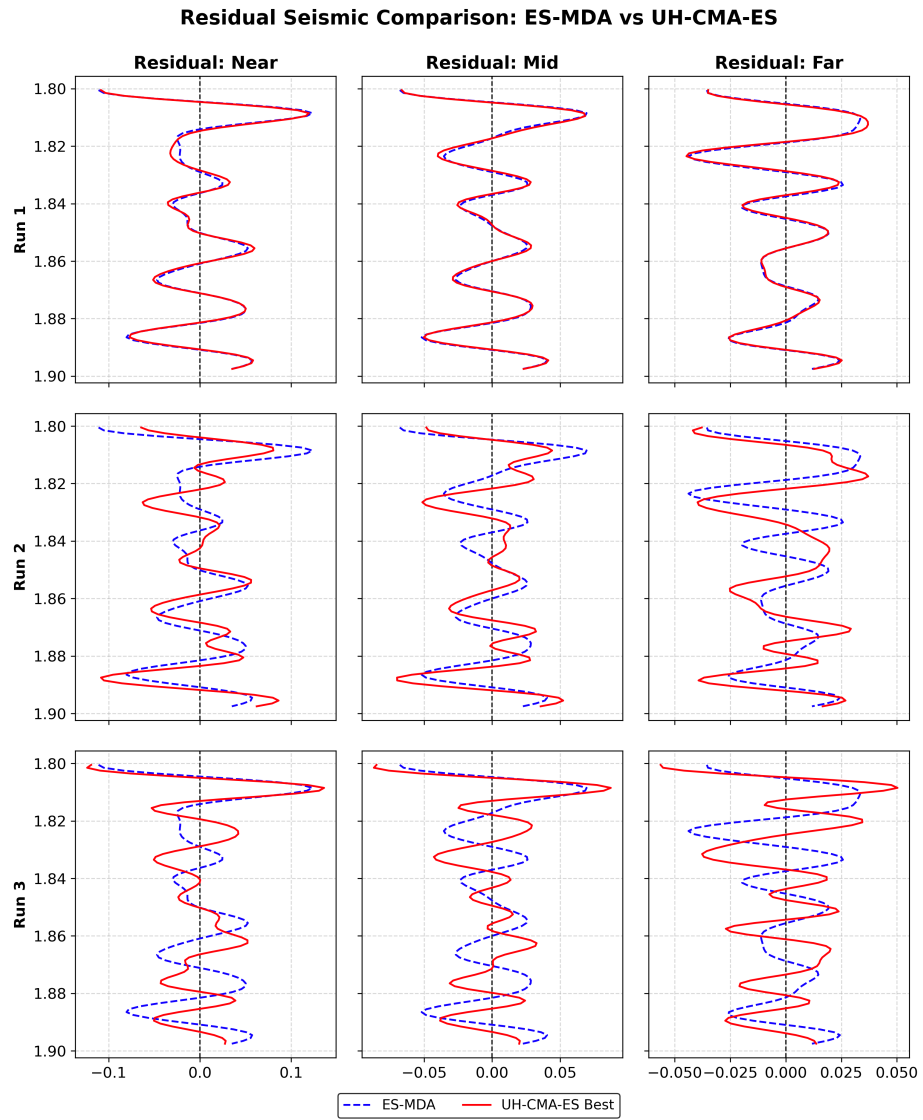


Figure 1: Comparison of simulated seismic residuals for Near, Mid, and Far incidence angles. The grid presents the performance of three independent runs of the evolutionary algorithm (rows) evaluated against the reference method. The curves illustrate the residual error for ES-MDA (solid blue line) and the best individual of UH-CMA-ES (solid red line).

Ensemble Comparison: ES-MDA vs UH-CMA-ES

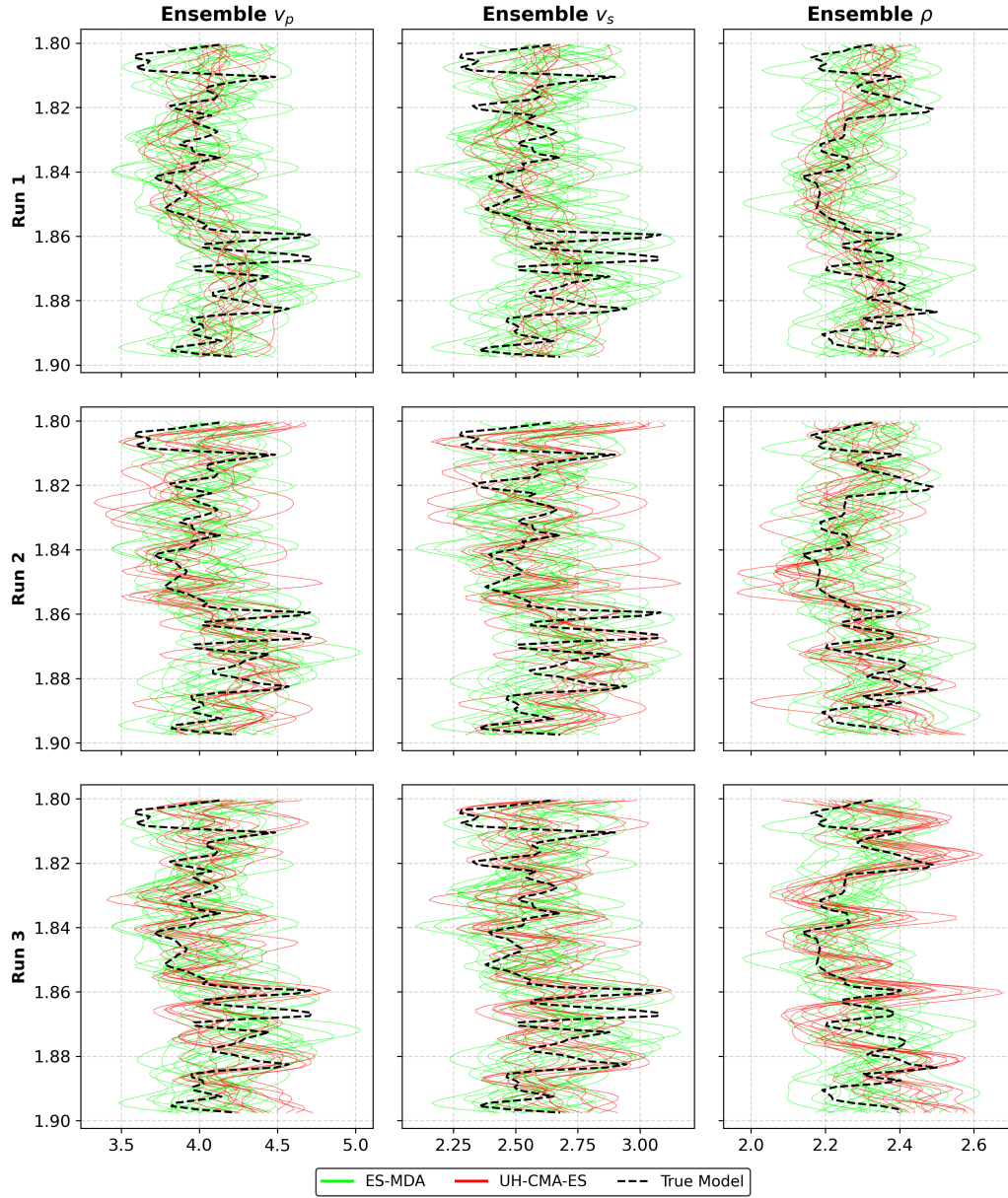


Figure 2: Depth (time) profiles of the estimated elastic properties: Compressional Velocity (v_p), Shear Velocity (v_s), and Density (ρ). The figure illustrates the spread of the ensembles for ES-MDA (in light green) and for three UH-CMA-ES configurations (in light red), overlaid on the ground-truth geological model (in black). The reduction in the uncertainty envelope (coverage area) achieved by the evolutionary method in its three runs compared to the global assimilation approach is visually observable.

To deepen the quantitative validation of the methods' accuracy, the non-parametric Kolmogorov - Smirnov (KS) test was applied, comparing the distributions of the mean estimates generated by each algorithm with the true model (ground truth). Table 1 summarizes the maximum distance statistics (KS Stat), the respective p -values, the 'spread' metric, that characterizes the uncertainty envelope's width, calculated as the standard deviation (or variance) of the estimated parameters across the ensemble, and the coverage for each inverted elastic property.

Table 1: Statistical metrics (Kolmogorov-Smirnov), spread, and ground-truth coverage comparing the reference (ES-MDA) and three independent runs of the UH-CMA-ES algorithm.

Method / Run	Metric	v_p	v_s	ρ
ES-MDA (Reference)	KS Stat	0.2755	0.3469	0.2347
	p -values	0.0008	< 0.0001	0.0073
	Spread	0.2376	0.1699	0.0894
	Coverage (%)	97.96	96.94	100.00
UH-CMA-ES (Run 1)	KS Stat	0.2551	0.2449	0.1939
	p -values	0.0026	0.0045	0.0435
	Spread	0.1791	0.1269	0.0688
	Coverage (%)	79.59	80.61	74.49
UH-CMA-ES (Run 2)	KS Stat	0.1429	0.2551	0.1837
	p -values	0.2498	0.0026	0.0644
	Spread	0.2107	0.1498	0.0732
	Coverage (%)	86.73	85.71	92.86
UH-CMA-ES (Run 3)	KS Stat	0.2143	0.2551	0.2347
	p -values	0.0186	0.0026	0.0073
	Spread	0.1317	0.0927	0.0472
	Coverage (%)	82.65	77.55	72.45

The analysis of Table 1 reveals that for the ES-MDA method and most UH-CMA-ES configurations, the p -values were below the standard significance level of 5% ($\alpha = 0.05$). This rejection of the null hypothesis of absolute equality between the distributions is the classic and expected behavior in the literature of ill-posed inverse problems; the inability to perfectly replicate the distribution of the true model evidences the complexity of non-uniqueness and the presence of noise in the observed data.

However, the major highlight of the proposed approach lies in the behavior of UH-CMA-ES Run 2. In this specific execution, the algorithm achieved p -values of 0.2498 for v_p and 0.0644 for ρ , both exceeding 0.05. Statistically, this implies the non-rejection of the null hypothesis, indicating that the distributions of these properties estimated by the evolutionary optimizer became mathematically indistinguishable from the true model (ground truth). This phenomenon proves that by concentrating exploration on a viable basin of attraction, UH-CMA-ES is capable of achieving a fit of exceptional statistical fidelity.

It is important to note that this higher precision entails a trade-off in coverage: while ES-MDA achieves coverage close to 100% due to its high spread, UH-CMA-ES presents values between 72% and 92%. Nevertheless, these coverage levels remain robust, demonstrating that the reduction in uncertainty promoted by the evolutionary method does not compromise the statistical representativeness of the true model, defining an uncertainty space that, although smaller in volume, is significantly more accurate relative to the target. As visually demonstrated in Figure 2, the uncertainty envelope generated by UH-CMA-ES (red lines) is significantly narrower and more focused

around the true model (black dashed line) compared to the broader global diversity maintained by ES-MDA (green lines).

The relative performance evaluation through the test statistic (KS Stat) in Table 1 also evidences advantages in the proposed approach. The KS Stat quantifies the maximum absolute distance between the cumulative distribution curves; smaller distances indicate greater proximity to the true model. For v_p and v_s , UH-CMA-ES showed significantly smaller distances compared to ES-MDA. This demonstrates that the evolutionary algorithm's focus on a specific basin of attraction, which reduced the global uncertainty area, translated into a statistically more faithful estimate of the actual subsurface properties.

Furthermore, it is important to note that the three independent runs of UH-CMA-ES differ solely in their initial random seeds. The superior statistical fidelity observed in Run 2 highlights the stochastic nature of the evolutionary process, where specific initial trajectories might navigate the non-linear objective function landscape more effectively toward the true model.

Finally, the computational cost, measured by the total number of forward simulations (N_{sim}), was 10^2 times higher for UH-CMA-ES compared to ES-MDA. The dynamic re-evaluation mechanism n_{eval} , necessary to ensure ranking stability under noise, makes the process more time-consuming compared to the direct and parallel update of ES-MDA. However, this greater computational effort is justified by obtaining a solution with narrower and more precise uncertainty bounds around the estimated parameters.

5. Conclusion

This work presented an evaluation of the UH-CMA-ES adaptive evolution algorithm in the context of seismic reservoir inversion, using the ES-MDA method as a benchmark. The results demonstrated that both methods are effective in minimizing seismic misfit, reaching equivalent convergence levels in the observed data domain. However, the main contribution of this study lies in the demonstration that UH-CMA-ES is capable of mapping the model space with significantly lower dispersion, generating more focused uncertainty envelopes that can be up to 45% narrower than those of the reference approach for the v_p property. Statistical analysis via the Kolmogorov-Smirnov test validated the qualitative superiority of the UH-CMA-ES local convergence, achieving distributions that were statistically indistinguishable from the ground truth in specific configurations.

Future research should focus on extending this methodology to the estimation of petrophysical properties, such as porosity, mineral volume fractions, and fluid saturation. This transition involves the integration of Rock Physics Models (RPM), which significantly increases the nonlinearity and multi-modality of the inverse problem compared to elastic inversion. To overcome the high computational cost (N_{sim}) observed in this study, investigations into architectural adaptations of the UH-CMA-ES are recommended. These include the implementation of high-performance parallel computing strategies to distribute population evaluations and the development of hybrid frameworks. Such hybrids could combine the global robustness of CMA-ES with surrogate-assisted optimization to reduce the number of required forward model calls, or incorporate local search refinements to further accelerate convergence in high-dimensional scenarios.

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